

Tangents to polar curves

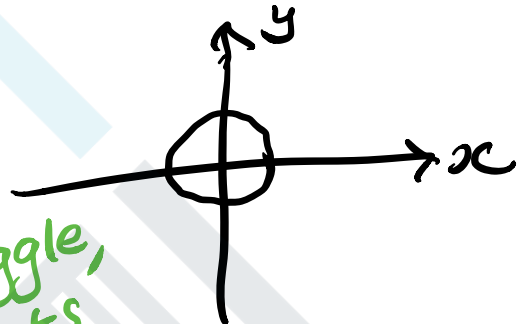
Basic skills

Q1. For $0 \leq \theta < 2\pi$

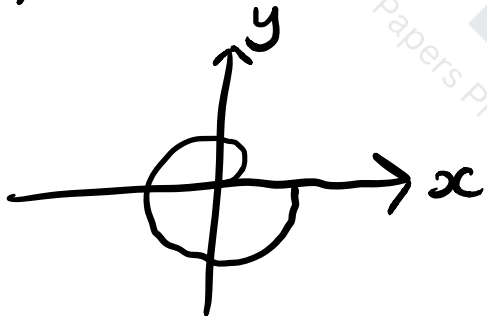
a) $r = \cos 3\theta$



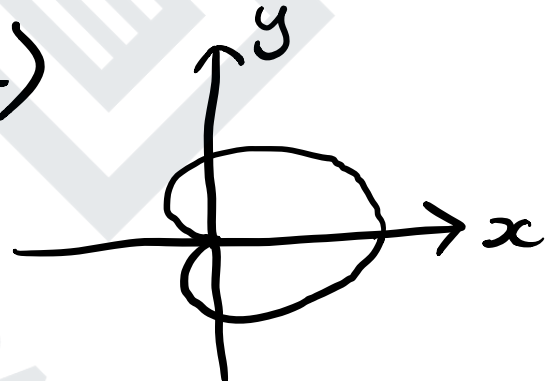
b) $r = 1$



c) $r = \theta$



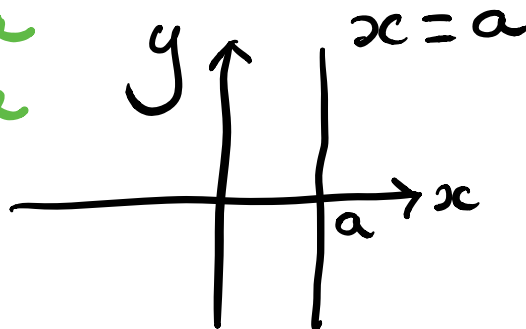
d)



If you struggle,
plot points

Q2. $r = a \sec \theta$

$r \cos \theta = a$
 $x = a$



$a \sec(b - \theta)$
has 'foot'
at (a, b)

Use addition
angle formula
to see this

Q3. a) $(x, y) = (-2, -4)$

$$r^2 = x^2 + y^2 = (-2)^2 + (-4)^2 = 4 + 16 = 20$$

$$r = \sqrt{20}$$

$$\theta = \tan^{-1}\left(\frac{-4}{-2}\right) = 1.11$$



$$\theta = \pi + 1.11$$

$$(r, \theta) = (\sqrt{20}, \pi + 1.11)$$

b) $(r, \theta) = (4, 5)$

$$x = r \cos \theta = 4 \cos 5 = 1.135$$

$$y = r \sin \theta = 4 \sin 5 = -3.836$$

$$(x, y) = (1.135, -3.836)$$

Q4. a) $\frac{d}{d\theta} (\overset{u}{\theta} \overset{v}{\sin \theta}) = \overset{\frac{du}{d\theta}}{1} \overset{v}{\sin \theta} + \overset{u}{\theta} \overset{\frac{dv}{d\theta}}{\cos \theta}$

Product rule

b) $\frac{d}{d\theta} (\sin(\overset{u}{\theta^2 + \pi})) = 2\theta \times \cos(\theta^2 + \pi)$

$\frac{du}{d\theta} = 2\theta$ $\frac{du}{d\theta} \times \frac{d}{du}(\sin u)$



Q5. $r = \cos \theta$

a) $\frac{dy}{d\theta} = \frac{d}{d\theta} (r \sin \theta) = \frac{d}{d\theta} (\cos \theta \sin \theta)$
 $= \frac{d}{d\theta} \left(\frac{1}{2} \sin 2\theta \right) = \frac{1}{2} \times 2 \cos 2\theta = \cos 2\theta$

double angle

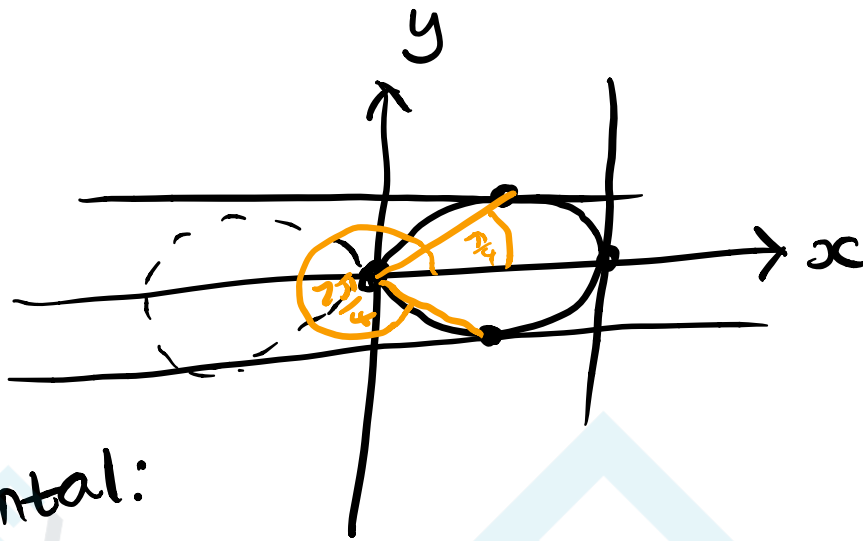
$\frac{dx}{d\theta} = \frac{d}{d\theta} (r \cos \theta) = \frac{d}{d\theta} (\cos^2 \theta)$
 $= -\sin \theta \cos \theta$

chain rule

for $\frac{dy}{d\theta} = 0$ solve $\cos 2\theta = 0$
 $\Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

$\frac{dx}{d\theta} = 0$ solve $-\sin \theta \cos \theta = 0$
 $\Rightarrow -\frac{1}{2} \sin 2\theta = 0$
 $\Rightarrow \sin 2\theta = 0$
 $\Rightarrow \theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$

b) $r = \cos \theta$



only 4 as
negative
none
existent
(1 same)

Horizontal:

$$\theta = \frac{\pi}{4} \Rightarrow r = \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}$$

$$\text{so } y = \frac{\sqrt{2}}{2} \sin \frac{\pi}{4} = \frac{1}{2}$$

by symmetry other horizontal is $y = -\frac{1}{2}$

↗ If unsure use $\theta = \frac{7\pi}{4}$

vertical:

$$\theta = 0 \Rightarrow r = 1 \quad \text{so } x = 1$$

$$\text{Other is } x = 0 \quad \& \quad \theta = \frac{\pi}{2}$$

Note $x = -1$ is only a tangent
to negative part

$$b) \quad r = 2(1 - \sin \theta), \quad 0 \leq \theta < 2\pi$$

$$\begin{aligned} \frac{dy}{d\theta} &= \frac{d}{d\theta} (2(1 - \sin \theta) \sin \theta) \\ &= \frac{d}{d\theta} (2 \sin \theta - 2 \sin^2 \theta) \end{aligned}$$

$$= 2 \cos \theta - 4 \sin \theta \cos \theta$$

$$\text{for } 2 \cos \theta (1 - 2 \sin \theta) = 0$$

$$\cos \theta = 0, \quad \sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2} \quad \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\text{So at } \theta = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\begin{aligned} \frac{dx}{d\theta} &= \frac{d}{d\theta} (2(1 - \sin \theta) \cos \theta) \\ &= \frac{d}{d\theta} (2 \cos \theta - 2 \sin \theta \cos \theta) \\ &= \frac{d}{d\theta} (2 \cos \theta - \sin 2\theta) \\ &= -2 \sin \theta - 2 \cos 2\theta \end{aligned}$$

for

$$-2\sin\theta - 2\cos 2\theta = 0$$

$$\sin\theta + \cos 2\theta = 0$$

$$\sin\theta + 1 - 2\sin^2\theta = 0$$

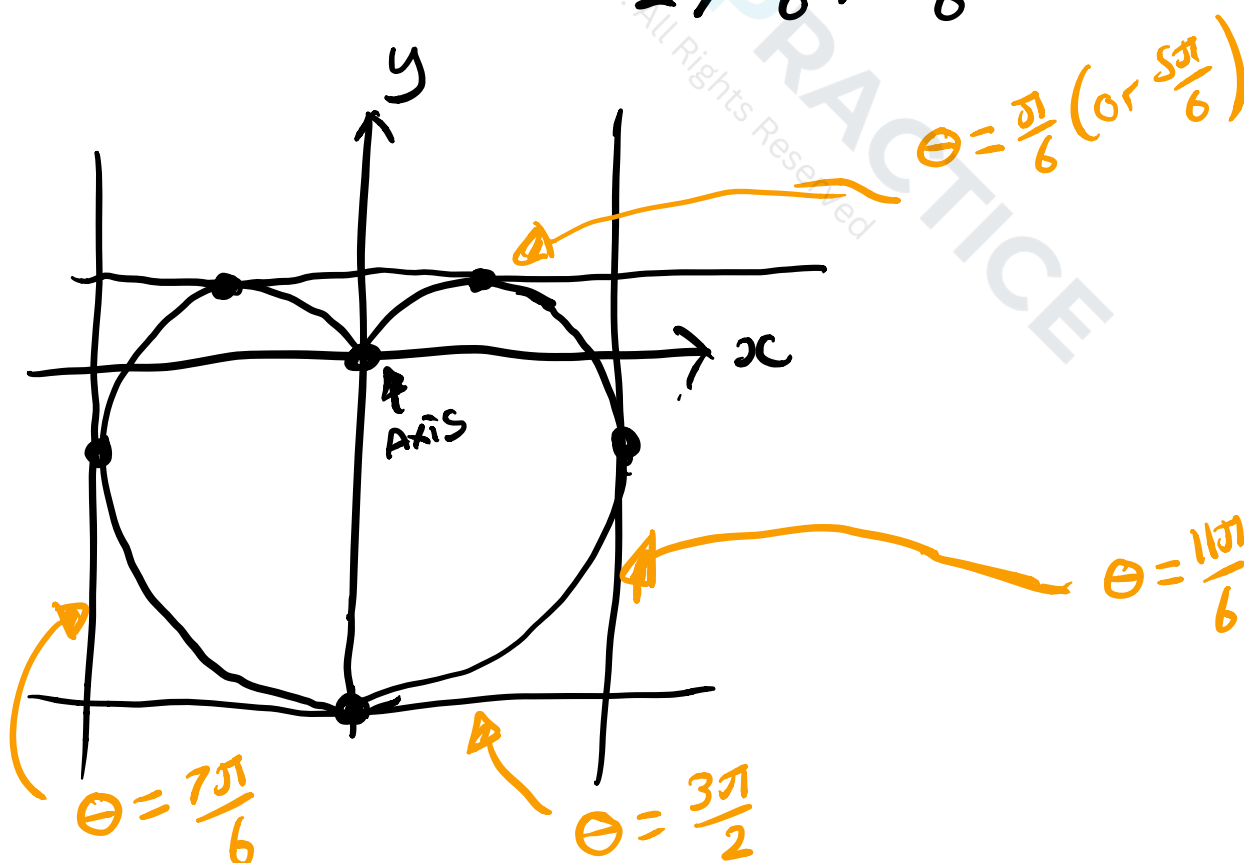
$$2\sin^2\theta - \sin\theta - 1 = 0$$

$$(2\sin\theta + 1)(\sin\theta - 1) = 0$$

$$\sin\theta = -\frac{1}{2} \quad \text{or} \quad \sin\theta = 1$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6} \quad \text{or} \quad \theta = \frac{\pi}{2}$$

$$\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$$



Horizontal

$$\text{At } \theta = \frac{\pi}{6} \quad r = 1$$

$$\text{so } y = 1 \times \sin \frac{\pi}{6} = \frac{1}{2}$$

$$\text{At } \theta = \frac{3\pi}{2} \quad r = 4$$

$$\text{so } y = 4 \times \sin \frac{3\pi}{2} = -4$$

vertical

$$\text{At } \theta = \frac{7\pi}{6} \quad r = 3$$

$$\text{so } x = 3 \times \cos \frac{7\pi}{6} = -\frac{3\sqrt{3}}{2}$$

$$\text{At } \theta = \frac{11\pi}{6} \quad \text{by symmetry}$$

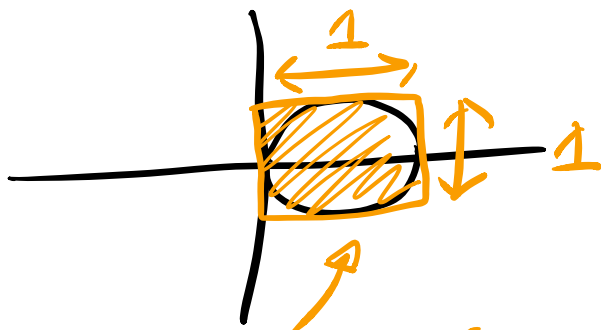
$$x = \frac{3\sqrt{3}}{2}$$

$$\text{Q7. } r = \sec \theta \quad \text{is} \quad r \cos \theta = 1$$

$$\Rightarrow x = 1$$

(This is the only tangent)

Q8. Area bounded by $r = \cos \theta$



Tangents (by Q5) are

$$x=1, x=0, y=\frac{1}{2}, y=-\frac{1}{2}$$

So Area $(1-0) \times (\frac{1}{2} - -\frac{1}{2}) = 1 \times 1 = 1$

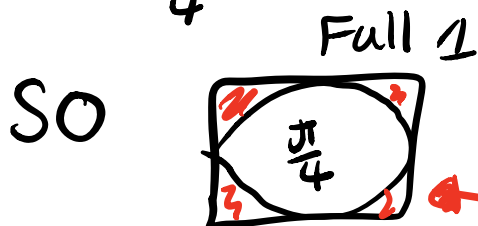
Advanced skills

Q9. Following Q8.

Area in curve $2 \times \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{2} + \frac{1}{2} \cos 2\theta d\theta = \left[\frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{4}$$



So $1 - \frac{\pi}{4}$ is outside
curve,
inside
rectangle

Q 10. $r = \sin \theta$

$$\frac{dy}{d\theta} = \frac{d}{d\theta} (\sin^2 \theta) = 2 \cos \theta \sin \theta$$

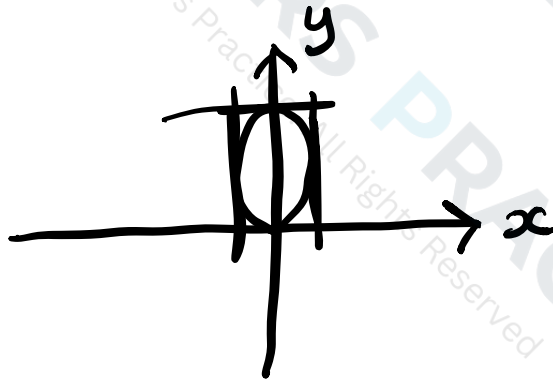
$$2 \cos \theta \sin \theta = 0 \Rightarrow \cos \theta = 0, \sin \theta = 0$$

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}, 0$$

$$\frac{dx}{d\theta} = \frac{d}{d\theta} (\sin \theta \cos \theta) = \frac{d}{d\theta} \left(\frac{1}{2} \sin 2\theta \right)$$

$$= \cos 2\theta$$

$$\cos 2\theta = 0 \Rightarrow \theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

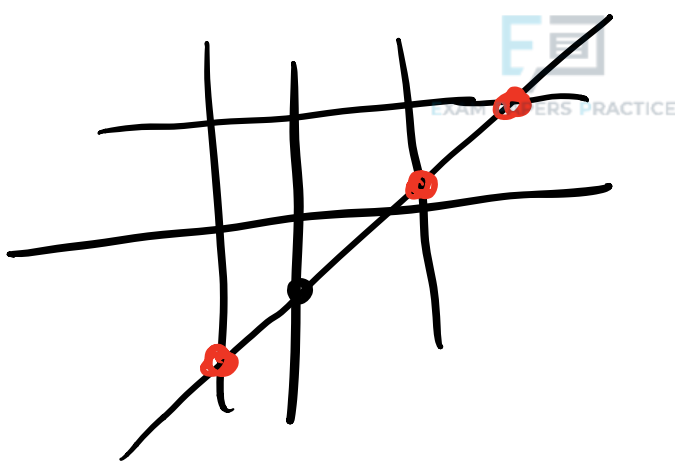


$$\theta = 0, \frac{\pi}{2} \Rightarrow y = 0, y = 1$$

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4} \Rightarrow x = \underbrace{\sin \frac{\pi}{4}}_r \underbrace{\cos \frac{\pi}{4}}_{\cos \theta} = \frac{1}{2}$$

and by symmetry

$$x = -\frac{1}{2}$$



$$y = 2x - 1 \text{ at } x = \frac{1}{2} \quad y = 0 \quad \left(\frac{1}{2}, 0\right)$$

$$x = -\frac{1}{2} \quad y = -2 \quad \left(-\frac{1}{2}, -2\right)$$

$$y = 1 \Rightarrow 2x - 1 = 1 \quad x = 1 \quad (1, 1)$$

$$\text{Q11. } r = \sqrt{6 \cos \theta}$$

$$\frac{dy}{d\theta} = \frac{d}{d\theta} \left(\sqrt{6} \sqrt{\cos \theta} \sin \theta \right)$$

$$= \sqrt{6} \left(\frac{-\sin \theta}{2\sqrt{\cos \theta}} \sin \theta + \sqrt{\cos \theta} \cos \theta \right)$$

$$= \sqrt{6} \left(\frac{-\sin^2 \theta}{2\sqrt{\cos \theta}} + \cos^{\frac{3}{2}} \theta \right)$$

$$\text{At } \theta = \frac{\pi}{4} \quad \frac{dy}{d\theta} = \sqrt{6} \times (-0.297 + 0.594)$$

$$= 0.121$$

$$\frac{dx}{d\theta} = \frac{d}{d\theta} \left(\sqrt{6} \cos^{\frac{3}{2}} \theta \right)$$

$$= \sqrt{6} \times -\sin \theta \times \frac{3}{2} \cos^{\frac{1}{2}} \theta$$

$$= -\frac{3\sqrt{6}}{2} \sin \theta \sqrt{\cos \theta}$$

$$\text{At } \theta = \frac{\pi}{4} \quad \frac{dx}{d\theta} = -2.184$$

$$\frac{dy}{dx} = \frac{0.121}{-2.184} = -0.0554$$

$$\frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

$$\text{Q12. } (x^2 + y^2)^{\frac{3}{2}} = 2xy$$

$$(r^2)^{\frac{3}{2}} = 2r^2 \sin \theta \cos \theta$$

$$r^3 = 2r^2 \sin \theta \cos \theta$$

$$r = \sin 2\theta$$

$$\frac{dy}{d\theta} = \frac{d}{d\theta} (\sin 2\theta \sin \theta)$$

$$= 2 \cos 2\theta \sin \theta + \sin 2\theta \cos \theta$$

$$2 \cos 2\theta \sin \theta + \sin 2\theta \cos \theta = 0$$

$$2(2\cos^2\theta - 1)\sin\theta + 2\sin\theta\cos^2\theta = 0$$

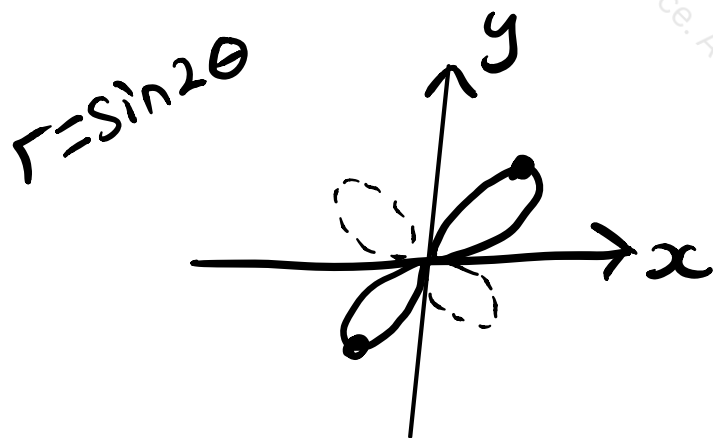
$$4\cos^2\theta\sin\theta - 2\sin\theta + 2\sin\theta\cos^2\theta = 0$$

$$2\sin\theta(3\cos^2\theta - 1) = 0$$

$$\sin\theta = 0 \quad \cos^2\theta = \frac{1}{3}$$

$$\cos\theta = \pm \frac{1}{\sqrt{3}}$$

$$\theta = 0, \pi$$



$$\theta = 0.955, \\ 5.329, \\ 2.186, \\ 4.097$$

$$\theta = 0.955 \text{ and } 4.097$$

$$x = \sin(2 \times 0.955) \cos(0.955) = 0.513$$

$$y = \sin(2 \times 0.955) \sin(0.955) = 0.791$$

$$(0.513, 0.791)$$

by symmetry ~~or~~ use $\theta = 4.097$
(Note. rounding
err will
compromise
Answers)
 $(-0.513, -0.711)$

